Renewable Energy Overview

Solar panels convert sunlight into electricity using photovoltaic cells.

Commercial silicon solar panels typically reach an efficiency between 18 and 22 percent.

Solar power works best in regions with high annual sunshine such as Rajasthan.

Batteries store the extra electricity generated during the day for use at night.

Wind and Hydropower

Wind turbines transform the kinetic energy of moving air into electrical energy.

Offshore wind farms produce more consistent power because sea winds are stronger and steadier.

Hydropower uses flowing water from dams to spin turbines and generate electricity.

Large dams can disrupt river ecosystems and displace local communities.